The Bombay Salesian Society's

Don Bosco Institute of Technology

(An Autonomous Institute Affiliated to the University of Mumbai)

CIRCUITS
CONNECT
COMMUNITIES

ENGINEERING
FOR A
BRIGHTER
TOMORROW

DBIT-EXTC SPECTRUM 2026

EXPLORE
ENGINEER
EMPOWER



WHAT'S INSIDE

- STUDENT'S ACHIEVEMENTS
- FACULTY CONTRIBUTIONS & ACHIEVEMENTS
- ACTIVITIES ORGANIZED BY DEPARTMENT

VOLUME 16

OUR PATRONS



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Rev. Fr. Nelson Carvalho

DEPARTMENT OF
ELECTRONICS AND TELECOMMUNICATION ENGINEERING

VISION

DBIT VISION

The Don Bosco Institute of Technology (DBIT) will be known to have an innovative, enjoyable and holistic learning environment that transforms individuals into socially conscious citizens the Don Bosco Way and will lead in research and entrepreneurship in the area of sustainable technologies.

DEPARTMENT VISION

To emerge as a leading department in the field of Electronics and Telecommunication Engineering, fostering holistic, ethical, and innovative professionals for sustainable societal development.

MISSION

DBIT MISSION

1. **To create future engineers who work with honesty and integrity and excel in the use of technology for the benefit of the underprivileged.**
2. **To train engineers to be innovative problem solvers and entrepreneurs who engage in research and life-long learning.**
3. **To provide a diverse and stimulating environment for staff and students to grow holistically.**

DEPARTMENT MISSION

1. **To produce technically competent, employable and socially responsible graduates with professional ethics.**
2. **To facilitate creativity, research, innovation, teamwork and entrepreneurial skills to meet academic, industrial and societal needs.**

Program Education Objectives (PEOs)

The graduates of the EXTC Program will be able to

- **Analyze, design and develop systems for applications in Electronics & Telecommunication and allied engineering fields with professional ethics.**
- **Pursue higher education in recognised national or international universities and carry out research & development work in the field of Electronics & Telecommunication Engineering.**
- **Get gainfully employed and/or become entrepreneurs in the field of EXTC Engineering.**

Program Specific Outcomes (PSOs)

PSO 1: Apply core concepts to analyze, design, and implement engineering solutions in Communication & Networking, VLSI & Embedded Systems, RF & Microwave, and Signal Processing domains.

PSO 2: Demonstrate industry-ready skills through project-based learning, using modern tools and/or open-source software to solve problems in Electronics & Telecommunication Engineering, while considering societal and sustainability aspects.

FROM THE PRINCIPAL'S DESK



Greetings!

It gives me great pleasure to present another edition of DBIT–EXTC SPECTRUM, reflecting the continued journey of learning, innovation, achievement, and engagement in the Department of Electronics and Telecommunication Engineering. Engineering education is evolving rapidly, driven by emerging technologies, changing industry expectations, and the growing need for interdisciplinary and sustainable solutions. In this context, our journey as an autonomous institution provides us with an important opportunity to make education more relevant, flexible, application-oriented, and responsive to the needs of industry and society.

I am happy to see the EXTC Department embracing this opportunity through academic initiatives, experiential learning, projects, industry and alumni interactions, technical activities, and skill development. These efforts encourage students to move beyond acquiring knowledge towards applying it meaningfully and developing the confidence and professional competencies required for their future careers.

I congratulate our students on their achievements and appreciate Dr. Madhavi Pednekar, Head of Dept. (EXTC), and all our faculty members for their commitment to mentoring, innovation, research, and holistic student development. I also acknowledge the valuable support of our alumni, parents, industry partners, and other stakeholders who continue to enrich our learning ecosystem.

My compliments to Ms. Pratibha Dumane, Editor of SPECTRUM, and the student editorial team for thoughtfully bringing together the department's accomplishments and experiences yet again in this edition.

As we move forward, I encourage our students and faculty to remain curious, adaptable, and responsible in their use of technology. I am confident that the EXTC Department will continue to grow, innovate, and contribute meaningfully to the academic journey of DBIT.

Wishing the EXTC Department continued progress and greater accomplishments ahead.

Dr. Sudhakar Mande
Principal

FROM THE HOD'S DESK



Greetings!

It is with immense pride and great pleasure that I present Volume 16 of SPECTRUM, the annual newsletter of the Department of Electronics and Telecommunication Engineering (EXTC).
This publication captures the milestones, achievements, and experiences that have enriched our department throughout the Academic Year 2025–26.
At EXTC, we remain committed to nurturing technically competent, socially responsible, and ethically grounded engineers—through interdisciplinary learning, industry interaction, expert lectures, hands-on lab experiences, and active engagement in our IEEE and IETE student chapters.
I extend my heartfelt gratitude to the DBIT Management and our respected Principal, Dr. Sudhakar Mande, for their unwavering encouragement and support in all our departmental initiatives.

This edition highlights our academic achievements, student projects, technical events, research, industry collaborations, and community engagement. Certificate courses, internships, workshops, industrial visits, and project-based learning continued to connect classroom learning with real-world practice.
I am particularly pleased with our students' achievements at Sankalpana 2026, Euphoria 2026, and Innovex 2026, along with our faculty's contributions to research and academic forums.
My sincere appreciation goes to the entire EXTC family—faculty, staff, students, alumni, and industry partners—for their valuable contributions to the department's growth. Special thanks to Prof. Pratibha Dumane, Newsletter Editor, the student editorial team, and Immanuel Machado for their efforts in bringing this edition to life.
As you turn these pages, I hope you gain a deeper appreciation of the achievements, aspirations, and vibrant culture that define our department.
Wishing you an engaging and inspiring read!

Dr. Madhavi Pednekar
Head of Dept. (EXTC)

FROM THE EDITOR'S DESK



It gives me immense pleasure to present yet another edition of DBIT–EXTC SPECTRUM, our newsletter that brings together the achievements, initiatives, experiences, and memorable moments that have shaped yet another eventful academic year.

The pages of this edition reflect the collective efforts of our students and faculty through academic achievements, projects, research, technical activities, industry and alumni interactions, workshops, internships, competitions, and other initiatives. Each contribution represents an opportunity to learn, participate, collaborate, and grow.

I sincerely thank the Management and our Principal, Dr. Sudhakar Mande, for their continued encouragement and support. I also acknowledge the Head of the Department, faculty members, staff, students, alumni, and everyone who has contributed to the activities featured in this edition. A special word of appreciation goes to the student editorial team — Pushkar Mahale, Swanandi Chavan, and Saksham More — for their creativity, commitment, and meticulous efforts in shaping and presenting this edition of SPECTRUM.

Having been associated with SPECTRUM for several years, it continues to be a rewarding experience to document the evolving story of the EXTC Department. As we look back at the milestones captured in these pages, I hope they inspire us to explore new possibilities and create many more achievements to celebrate in the editions ahead.

Ms. Pratibha Dumane
Dean Academics,
Faculty Editor, Spectrum

Students' Academic Achievements (2024-25)



S.E. EXTC Semester III

1	Jaiswal Yukita Jagdish Mamta
2	Mediboina Pritika Krishnarao Rohini
3	Sharma Raman Lalchand Kavita

S.E. EXTC Semester IV

1	Mediboina Pritika Krishnarao Rohini
2	Sharma Raman Lalchand Kavita
3	Jaiswal Yukita Jagdish Mamta

T.E. EXTC Semester V

1	Pawar Arpita Avdhut Anjali
2	Shaikh Aftab Alam
3	Belgaonkar Akash Vilas Pallavi

T.E. EXTC Semester VI

1	Belgaonkar Akash Vilas Pallavi
2	Pawar Arpita Avdhut Anjali
3	Shaikh Aftab Alam



B.E. EXTC Semester VII

1	Rai Diya
2	Garje Yash
3	Annanya Zadbuke

B.E. EXTC Semester VIII

1	Garje Yash
2	Rai Diya
3	Annanya Zadbuke

Students' Co-Curricular Achievements



Swayam Dalvi (BE) underwent the following certification courses offered by Udemy:

- Machine Learning A-Z: AI, Python & R + ChatGPT Prize [2026].
- Artificial Intelligence A-Z 2026: Agentic AI, Gen AI, and RL.
- TinkerCAD for Complete Beginners

Swayam Dalvi completed a 15-day training programme on "Fundamentals of Mechatronics and Automation" at the Indian Institute of Skills, Mumbai.

Saarth Gandre (SE) completed "Software Engineering Job Simulation" offered by Forage, Accenture from January-August 2025.

Arya Munde completed the following certificate courses offered by Udemy:

- Python for Data Science and Machine Learning Bootcamp
- The Complete SQL Bootcamp: Go from Zero to Hero

Ashlesha Tambat (SE) participated in Quasar 4.0: 24 Hrs National Level Hackathon of Unstop Holiday Fest 2025 organised by Vasantdada Patil Pratishthan's College of Engineering and Visual Arts, Mumbai.

Ashlesha Tambat completed the following certificate courses offered by Udemy:

- Advanced C Programming Course
- The Complete SQL Bootcamp: Go from Zero to Hero
- Microcontroller Embedded C Programming: Absolute Beginners

Samiksha Narayan Gaonkar (TE) completed the following courses:

- 3-week Virtual Internship on "Embedded Systems, IoT & PCB Designing" from 11 August 2025 organised by IIIRPD, Hyderabad in technical co-sponsorship with IEEE Robotics and Automation Society Hyderabad Section Chapter & IEEE Communication Society Hyderabad Section Chapter.
- Three weeks program in "Fundamentals of Mechatronics & Automation" offered by TATA Indian Institute of Skills.

Students' Co-Curricular Achievements

Mangesh Kanse (TE) attended the following:

- Masterclass – Python Basics to Brilliance, conducted by Ethan's Tech Solutions, LLP on 27 April 2025.
- 20-hour course on "artificial Intelligence & Machine Learning" in a 4-week internship on Emerging Technologies from 20 January 2026 – 12 February 2026, a Microsoft Elevate Initiative in collaboration with the All India Council for Technical Education (AICTE).

Zubia Sarang (BE) completed summer internship on "Data Science & Deep Learning" organised by Dept. of AI & DS, VCET & IIIC, VCET in association with Rogue code from 16 – 27 June 2025.

Zubia Sarang (BE) and Moin Khan (TE) secured the 1st position in the Manipal Hackathon 2025 organised by Manipal Institute of Technology from 8-10 October 2025.

Amey Kanade completed training and internship with SmartED Innovations in "Very Large Scale Integration" in December 2025.

Kapil Labde (TE) completed the Lab Workshop on "Semiconductor Devices – From Microscopic Control to Real-World Systems" conducted by Dept. of Electrical Engineering, IIT Bombay from 22-27 June 2026.



Students' Extra-Curricular Achievements

Sr. No.	Event / Competition	Category	Participant / Winner Name(s)	Position / Prize
1	Solo Dance (Western)	Cultural	Aarya Jadhav (SE)	1st Prize
2	Solo Dance Battle	Cultural	Aarya Jadhav (SE)	3rd Place
3	Group Dance	Cultural	Aarya Jadhav (SE) Yusra Ansari (SE) Swanandi Chavan (SE)	3rd Place
4	Solo Singing	Cultural	Sumit Botle (TE)	3rd Place
5	Group Debate	Literary	Jashith Shinde (TE)	2nd Place
6	Running 400m (Girls)	Sports	Ocina Serrao (SE)	2nd Prize
7	Running 200m (Girls)	Sports	Ocina Serrao (SE)	3rd Prize
8	Running 80m (Girls)	Sports	Mrunmai Paktekar (TE)	3rd Prize
9	4x200m Girls Relay	Sports	Mrunmai Paktekar (TE) Ocina Serrao (SE)	3rd Place
10	Table Tennis (Duo)	Sports	Ocina Serrao (SE)	3rd Prize
11	Table Tennis		Siya Agivale (SE)	2nd Place
12	5.5km Marathon	Sports	Sakshi Salunkhe (SE)	3rd Place
13	Volleyball	Sports	Aarya Jadhav (SE) Mrunmai Paktekar (TE) Yusra Ansari (SE) Swanandi Chavan (SE) Siya Agivale (SE)	2nd Place
14	Box Cricket	Sports	Sakshi Salunkhe (SE) Aarya Jadhav (SE) Mrunmai Paktekar (TE) Yusra Ansari (SE) Siya Agivale (SE)	2nd Place
15	4x200m Relay (EXTC Girls)	Sports	Ocina Serrao (SE) Mrunmai Paktekar (TE)	3rd Prize

Students' Extra-Curricular Achievements



Swanandi Chavan and Ocina Serrao (SE) were members of the team that secured the 2nd Place in the Maschio Cup Girls Throwball Event at Don Bosco, Matunga on 30 November 2025.



Ocina Serrao (SE), Siya Agivale (SE), Swanandi Chavan (SE), Mrunmai Paktekar (TE) were members of the team that secured the 1st place in the Sportsmania Girls Throwball Event organised at Vidyalankar Institute of Technology, Mumbai on 21 February 2026.

Students' Extra-Curricular Achievements



Amit Patil (SE) was a member of the DBIT Kabaddi team at the University of Mumbai Kabaddi Tournament on 18 September 2025.



Amit Patil (SE) was a member of the PK Warriors Team that secured the 1st position in DBPL 2026 Event organised on 02 May 2026.

Guest Lectures Organised for Students

Sr. No.	Topic	Resource Person	Organisation	Organised By	Date
1	Neural Networks in Action: Emerging Trends and Industry Insights	Mr. Bilal Hungund	Sr. Data Scientist, Halliburton (DBIT IT Alumnus - Batch of 2019)	Ms. Pratibha Dumane	13 Apr 2026
2	AI, Semiconductors and You: Positioning for the Next Computer Decade	Mr. Deepak Tanna	Principal Engineer, Intel India Technology Private Ltd., Bengaluru.	Dr. S. S. Mande	20 Apr 2026
3	Design and Verification Talent for Semi-conductor Technology	Mr. Kamlesh Mulchandani	Semiconductor Start Up Professional and Hands-on Trainer in VLSI. (DBIT EXTC Alumnus - Batch 2011)	Dr. S. S. Mande	6 May 2026

Industrial Visits

A two-day industrial visit to BSNL ZTTC and GMRT, Pune was organized on 27-[illegible] February 2026 for the Third-Year students. The visit was coordinated by Dr. Aparna Telgote and accompanied by Dr. Royal D'Souza and Mr. Deepak Khake.





An industrial visit to BSNL Earth Station, Yeoor, Thane was organized for Final Year EXTC students. The visit was coordinated by Ms. Poonam Chakraborty & Dr. Aparna Telgote.





Industrial Visits

On 26 September 2025, 57 students visited tribal villages near Medhe, Vasai Tehsil, Palghar district, facilitated by the Aga Khan Agency for Habitat India (AKAHI).
The faculty coordinators were Dr. Ashwini Kotrashetti and Ms. Freda Carvalho.



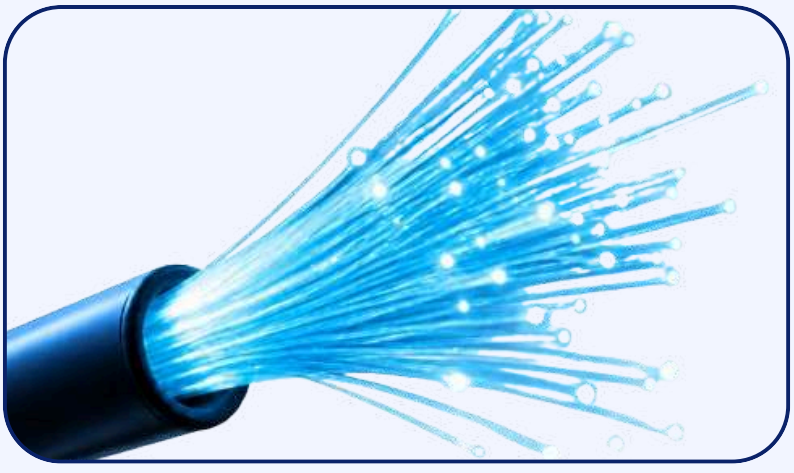
Fiber Optic Splicing Workshop





A hands-on workshop on “Fiber Optic Communication” by Dr. Nilesh Nikam, Director, eAmplify, Mumbai, was organized on 23 March 2026 for the BE EXTC students by Ms. Poonam Chakraborty.

The workshop provided practical exposure to fusion splicing and OTDR, helping students understandfibre optic network installation, testing, fault detection,and performance analysis, while enhancing their industry readiness.



InnoQuest
Mini Project Exhibition & Competition







InnoQuest, a Mini Project Exhibition and Competition on 31 October 2025, provided an excellent platform for Second-Year (SE) and Third-Year (TE) students to showcase their innovative projects across diverse domains such as IoT, Embedded Systems, and Communication.

The event witnessed enthusiastic participation, with a total of 143 students presenting 37 projects, reflecting creativity, technical proficiency, and practical application of engineering concepts.

The top-performing teams were recognized and awarded for their innovation, technical excellence, and effective presentation skills.

The event was coordinated by Ms. Freda Carvalho, SE Mini Project Coordinator and Dr. Aparna M. Telgote, TE Mini Project Coordinator.

InnoQuest Winners from S.E.

Rank	Title of the Project	Team Members
I	Smart Density Based Traffic Signal Management System	1. Giri Sapnakumari Ganesh 2. Haldankar Neha Nitin 3. Hasan Mohammed Musham 4. Kanade Amey Prashant
II	Smart Irrigation	1. Khanvilkar Yash Rajesh 2. Kumar Rohit Anil Devmuni 3. Patil Harshit Apekshit 4. Shetty Tirth Thimmappa
III	Medicine Reminder & Monitoring Device for Elderly	1. Kenneth Dsouza 2. Mishra Disha Ratnesh 3. Raj Vats 4. Sneha Satish Savekar

InnoQuest Winners from T.E.

Rank	Title of the Project	Team Members
I	IoT Based Smart Inhaler	1. Mrunmai Paktekar 2. Kripa Dewoolkar 3. Aryan Puranik 4. Saeesh Gidh
II	Care-Talk	1. Yashvi Jain 2. Anu Saha 3. Dhruv Tare 4. Vipaak Gaikwad
III	IoT-Enabled Weather Station using ESP32	1. Nabeela Ansari 2. Prathamesh Yerekar 3. Vishakha Jamnekar 4. Sushant Patil



Board of Studies (BoS) Meeting

The Board of Studies (BoS) meeting of the Department of Electronics and Telecommunication Engineering (EXTC), DBIT was conducted on 23 June 2026.

The BoS members included Dr. Mukesh Patil – Principal, Ramrao Adik Institute of Technology, Navi Mumbai, Dr. Milind Shah – Professor, Fr. C. Rodrigues Institute of Technology, Navi Mumbai, Mr. Gaurav Pednekar – Lead Engineer, Lear Corporation, Mr. Manjesh Salian - Capgemini Technology Services India Ltd. and Dr. Kishor Sawardekar – Associate Professor, IIT-BHU.

The meeting brought together academic and industry experts to review and discuss the curriculum, syllabus, examination structure, and academic initiatives of the EXTC Engineering Programme under autonomy.



Parent-Teacher Meetings

The Parent–Teacher Meetings were conducted on 11 October 2025 and 20 June 2026 to facilitate effective communication between faculty members and parents regarding students' academic performance, attendance, discipline, and overall development.

The meetings provided a platform for parents to interact with the Head of the Department and faculty members, discuss students' strengths and areas for improvement, and understand the academic and professional development initiatives undertaken by the department.

The interaction strengthened the collaboration between parents and the department and encouraged a joint effort towards the holistic development of the students.







Internships

Sr. No.	Student Name	Class	Organisation	Duration
1	Prerna Gawale	BE	Jacob Solution	01 June - 31 July 2025
2	Om Hindalkar	BE	Robotic Intern	21 April - 31 July 2025
3	Sarvambh Desai	BE	Mastek Quad Gen	23 June - 22 August 2025
4	Aditya Shetty	BE	Akasa Air	09 June - 08 August 2025
5	Jasmit Ganvir	TE	TATA Power, Mumbai	12 - 17 January 2026
6	Priyanshu Sakharkar	TE	TATA Power, Mumbai	12 - 17 January 2026
7	Soham Derdekar	TE	TATA Power, Mumbai	12 - 17 January 2026
8	Omkar Dhebe	TE	TATA Power, Mumbai	12 - 17 January 2026
9	Jero Royston	BE	Burns & McDonnell	5 January - 4 June 2026
10	Prerana Gawale	BE	10Cubits Pvt. Ltd.	5 January - 4 June 2026
11	Soham Ghadigaonkar	BE	Selec Control Ltd.	5 January - 4 June 2026
12	Prathamesh Kurdekar	BE	Selec Control Ltd.	5 January - 4 June 2026
13	Samruddhi Patil	BE	Selec Control Ltd.	5 January - 4 June 2026
14	Arpita Pawar	BE	Selec Control Ltd.	5 January - 4 June 2026
15	Aditya Punekar	BE	Selec Control Ltd.	5 January - 4 June 2026

Internships

16	Aakash Belgaonkar	BE	Selec Control Ltd.	5 January - 4 June 2026
17	Prerana Gawale	BE	Econship Tech Pvt. Ltd.	12 January - 16 February 2026
18	Sumit Shinde	BE	AV Nirvana Distribution	04 March - 02 June 2026
19	Swayam Dalvi	BE	IIDE	16 February - 04 June 2026
20	Anuskha Tewar	SE	CrEAST (L&T- DBIT)	01 December 2025 - 28 February 2026
21	Athiya Shetty	SE	CrEAST (L&T- DBIT)	01 July 2025 - 28 February 2026
22	Swara Ghosalkar	SE	CrEAST (L&T- DBIT)	01 July 2025 - 28 February 2026
23	Kapil Labde	TE	CrEAST (L&T- DBIT)	01 July 2025 - 28 February 2026
24	Aayush Jangam	TE	CrEAST (L&T- DBIT)	01 July 2025 - 28 February 2026
25	Nirmiti Bait	TE	CrEAST (L&T- DBIT)	01 July 2025 - 28 February 2026
26	Kartik Dandelia	TE	CrEAST (L&T- DBIT)	01 July 2025 - 28 February 2026
27	Pratham Ghagre	TE	CrEAST (L&T- DBIT)	01 July 2025 - 28 February 2026
28	Jain Yashvi	TE	CrEAST (L&T- DBIT)	01 July 2025 - 28 February 2026
29	Dhruv Tare	TE	CrEAST (L&T- DBIT)	01 July 2025 - 28 February 2026
30	Pritika Mediboina	TE	CrEAST (L&T- DBIT)	01 July 2025 - 28 February 2026

Internships

31	Sneha Pawar	TE	CrEAST (L&T- DBIT)	01 July 2025 - 28 February 2026
32	Harshali Gokhale	TE	CrEAST (L&T- DBIT)	01 July 2025 - 28 February 2026
33	Kartik Nyalapelli	TE	CrEAST (L&T- DBIT)	01 July 2025 - 28 February 2026
34	Mascarenhas Blossom	TE	CrEAST (L&T- DBIT)	01 July 2025 - 28 February 2026
35	Shejwal Janhavi	TE	CrEAST (L&T- DBIT)	01 July 2025 - 28 February 2026
36	Dewoolkar Kripa	TE	CrEAST (L&T- DBIT)	01 July 2025 - 28 February 2026
37	Anushka Tewar	SE	CrEAST (L&T- DBIT)	01 July 2025 - 28 February 2026
38	Samiksha Gaonkar	TE	CrEAST (L&T- DBIT)	01 July 2025 - 28 February 2026
39	Daniel Sebastian	TE	CrEAST (L&T- DBIT)	01 July 2025 - 28 February 2026
40	Yukita Jaiswal	TE	CrEAST (L&T- DBIT)	01 July 2025 - 28 February 2026
41	Samiksha Gaonkar	TE	IIIRPD, Hyderabad	11 - 29 August 2025
42	Christoval Dsouza	TE	Hindustan Aeronautics Ltd. Bengaluru	15 June - 15 July 2026

Synergy 2026

Talent , Triumph & Togetherness

The Department Day, Synergy was celebrated on 27 April 2026, with “Red Carpet” as the theme for this year. The event witnessed enthusiastic participation from around 250 students across FE, SE, TE, and BE.

The event featured a vibrant mix of cultural performances, including dance, music, fashion show, and drama, showcasing the creativity and talent of students and faculty.

Academic and co-curricular achievements were also recognized with student felicitations. Coordinated by Dr. Aparna Telgote, the celebration successfully fostered teamwork and community, concluding with a lively jam session.







Journal Publication By Faculty

1. **Carvalho Freda**, Kotrashetti Ashwini, Bhattacharyya Kaustubh, “Mode Analysis for Prediction of Heating Patterns in Microwave Cavities Powered by Magnetron”, published on 11 March 2026 in Progress in Electromagnetics Research C, Vol. 167, 39-49, 2026. doi: 10.2528/PIERC26011705
2. **Carvalho Freda**, Kotrashetti Ashwini, Bhattacharyya Kaustubh, “Investigation of Heating in Microwave Cavities: Insights, Evolution, and Advancements”, published on 13 October 2025 in IETE Journal of Research, pp. 1–16, 2025. doi:10.1080/03772063.2025.2567602
3. **Sonal Patil**, Ashwini Naik, “Examination of Breast Phantom Layers Using UWB Antenna Sensing Technique”, International Journal of Electrical and Computer Engineering Systems, Volume 17, Number 4, 2026. doi: 10.32985/ijeces.17.4.
4. Sagar Mhatre, Vinitkumar Dongre, **Sudhakar Mande**, “A Hardware-Software Co-Design Framework for Energy Efficient LSTM Based SoC Estimation on Resource-Constrained FPGAs”, J. Comput. Mech. Manag., vol. 5, no. 3, pp. 157–174, Jun. 2026.
5. Sagar Mhatre, Vinitkumar Dongre, **Sudhakar Mande**, “MIRA: Memory-Centric Intelligent Reconfigurable Architecture for LSTM”, International Journal of Electrical and Electronics Engineering, vol. 13, no. 5, pp. 127-142, 2026. https://doi.org/10.14445/23488379/IJEEE-V13I5P111.

Conference Papers

1. **Dibyarupa Pradhan, Yash Garje, Ananya Boliyar, Sakshi Shidruk,** Ashwini Kotrashetti, **Freda Carvalho**, “Design and Evaluation of Four Antenna Prototypes: Selection of an Optimal and Miniaturized Antenna for Anti Drone Sub System”, in INDISCON 6th International Conference, Aug. 2025.

Conference Papers

2. **Poonam Chakraborty**, Anudeepa Kholapure, Rupali Patil, “A Comprehensive Review of Blockchain-Based Security Solutions in Mobile Adhoc Networks”, IEEE Xplore, Proceedings of ICETI4T 2025. doi: 10.1109/ICETI4T63625.2025.11132175.

3. Sagar Mhatre, Vinitkumar Dongre, **Sudhakar Mande**, "Efficient Deep Learning Inference Using a Parameterizable LSTM Accelerator on FPGA," 2025 International Conference on Data, Energy and Communication Networks (DECoN), Bhopal, India, 2025, pp. 1-6, doi: 10.1109/DECoN67170.2025.11447718.

4. Sagar Mhatre, Vinitkumar Dongre, **Sudhakar Mande**, "Enhancing Edge Performance: A Comparative Analysis of LSTM Inference Using Hardware Acceleration," 2025 International Conference on Machine Learning and Autonomous Systems (ICMLAS), Prawet, Thailand, 2025, pp. 1569-1574, doi: 10.1109/ICMLAS64557.2025.10968872.

Design Registration

1. Design registration for “C Band Metasurface Filter” - Design No. 461993- 001 issued on 21 October 2025 in the name of 1. Don Bosco Institute of Technology 2. Dr. Ashwini Kotrashetti 3. Freda Carvalho 4. Ansari Mohd Raza 5. Aditya Jadhav.
2. Design registration for “All Metal High Power Vivaldi” - Design No. 468033-001 issued on 17 June 2026 in the name of 1. Don Bosco Institute of Technology 2. Dr. Ashwini Kotrashetti 3. Freda Carvalho 4. Amey Dilip Bhombade 5. Diya Rai 6. Annanya Zadbuke 7. Rohit Mali.
3. Design registration for “Method and system for IoT Skill Development using Dual Microcontrollers and Comprehensive Sensor Integration” in the name of 1. Sudhakar Mande 2. Dipak Pandav.

Sankalpana 2026

SANKALPANA 2026, a Mini Project Exhibition and Competition, on 14 May 2026, was organized by the Department and the IETE-DBIT-ISF. The event provided a platform for SE and TE students to showcase innovative projects in embedded systems, IoT, automation, digital electronics, and communication systems, fostering innovation, teamwork, and practical engineering skills.

WINNERS

Rank	Project Title	Team Members
1	Traffic Signal Management System	1. Sapnakumari Giri 2. Neha Haldankar 3. Musham Hasan 4. Amey Kanade
2	Smart Irrigation	1. Yash Khanvilkar 2. Rohit Kumar 3. Harshit Patil 4. Tirth Shetty
3	Medicine Reminder and Monitoring Device for Elderly	1. Kenneth Dsouza 2. Disha Mishra 3. Raj Vats 4. Sneha Savekar



Sankalpana 2026



RANK 1

RANK 2





RANK 3

Placements AY 2025-26



Sr. No.	Name of Student	Company
1	Royston Poobalarayan	Burns & McDonnell
2	Priyanshu Sakharkar	Arcon Pvt. Ltd.
3	Neel Pujari	Capgemini
4	Aditya Shetty	Capgemini
5	Ayush Gajbhiye	2Factor
6	Tanmayee Ahire	Godrej and Boyce
7	Neel Pujari	Godrej and Boyce
8	Soham Derdekar	Godrej and Boyce
9	Aditya Shetty	Godrej and Boyce
10	Prerana Gawale	Econship
11	Soham Ghadigaonkar	Selec Control Ltd.
12	Samruddhi Patil	Selec Control Ltd.

13	Prathamesh Kurdekar	Selec Control Ltd.
14	Akash Belgaonkar	Selec Control Ltd.
15	Arpita Pawar	Selec Control Ltd.
16	Gopal Jha	Selec Control Ltd.
17	Aditya Punekar	Selec Control Ltd.
18	Akash Belgaonkar	QSpider
19	Prathamesh Patil	QSpider
20	Sarvambh Desai	QSpider
21	Arya Munde	QSpider
22	Jasmit Ganvir	Selec Control Ltd.
23	Swayam Dalvi	Selec Control Ltd.
24	Narayan Dalvi	Selec Control Ltd.
25	Prathamesh Patil	Selec Control Ltd.
26	Sumit Shinde	AV Nirvana
27	Jaden Fernandes	AV Nirvana
28	Prince Kumar	InternsElite
29	Aditi Mishra	InternsElite
30	Nisha Singh	RSM Astute Pvt. Ltd.
31	Prerna Gawale	RSM Astute Pvt. Ltd.
32	Arya Munde	Reliance Retail
33	Narayan Dalvi	MidGate Pvt. Ltd.

Innovex 2026

IINNOVEX 2026 served as a dynamic platform for final-year students to showcase their major projects, demonstrating technical creativity and problem-solving skills across emerging technologies.
The event allowed students to present their real-world solutions directly to experts and gain valuable industry feedback.

Rank	Project Title	Student Team	Project Guide
1	AI-based Interview Simulator with Real-Time Response Analysis	1. Harsh Dusane 2. Aakash Belgaonkar 3.Samruddhi Kapadnis 4. Ayush Gajbhiye	Ms. Pratibha Dumane
2	Design & Development of High-Power Vivaldi Antenna	1. Arpita Pawar 2. Samruddhi Patil 3. Aftab Shaikh 4. Jero Royston	Ms. Freda Carvalho
3	Environmental Monitoring Station	1.Prathamesh Kurdekar 2. Neel Pujari 3. Nisha Singh 4.Priyanshukumar Singh	Dr. Madhavi Pednekar Mr. Prasad Padalkar (IT Dept.)
D. Sundararajan Award	AI Based Cybersecurity System	1. Priyanshu Sakharkar 2. Yashi Nimje 3. Soham Ghadigaonkar 4. Tanmayee Ahire	Ms. Poonam C.

Innovex 2026 EXTC Winners



RANK 1



RANK 2



RANK 3



D. Sundararajan Award

Proud Milestones Ph.D. Achievements

1. **Dr. Sudhakar Mande** guided five candidates under the University Mumbai to the successful completion of their Ph.D. degrees during AY 2025–26.
2. **Dr. Aparna Telgote**, Assistant Professor, completed Ph.D. degree from University of Mumbai under the guidance of Dr. Sudhakar Mande in AY 2025-26.
3. **Dr. Freda Carvalho**, Assistant Professor, Department of Electronics and Telecommunication, successfully completed the Ph.D. degree from Assam Don Bosco University in June 2026.
4. **Dr. Sonal Patil,** Assistant Professor, Department of Electronics and Telecommunication, successfully completed the Ph.D. degree from D. Y. Patil University, Navi Mumbai, on 09 April 2026.
5. **Dr. Shilpa Gaikwad**, Assistant Professor, Department of Electronics and Telecommunication, successfully completed the Ph.D. degree from Symbiosis International (Deemed University), Pune on 27 April 2026.



Dr. Freda Carvalho



Dr. Aparna Telgote



Dr. Sonal Patil



Dr. Shilpa Gaikwad

Faculty as Resource Persons

- **Ms. Pratibha Dumane** delivered a talk on “Research & Publication Ethics”, organised by IETE-ETSA Student's Forum for the EXTC students at Fr. C. Rodrigues Institute of Technology, Vashi, Navi Mumbai, on 16 October 2025.
- **Dr. Sudhakar Mande** delivered a talk on “Operational Transconductance Amplifier (OTA) Design” as part of a Five-Day Faculty Development Programme (FDP) under the C2S – Chip to Startup Initiative (supported by MeitY and CHIPIN) on 03 December 2025.
- **Dr. Sudhakar Mande** delivered an expert talk on “Low Power CMOS VLSI Design” during ATAL FDP on Periodic innovations in perspective of VLSI of Semiconductor ICs organized by Shah & Anchor Kutchhi Engineering College, Mumbai on 20 December 2025.
- **Ms. Pratibha Dumane** delivered a talk on “Time Management and Stress Management” in the Faculty Development Program on “Empowering Faculty for Holistic Professional Growth”, organised by Department of Basic Sciences & Humanities in collaboration with Dept. of Electrical Engineering at Fr. C. Rodrigues Institute of Technology, Vashi, Navi Mumbai on 06 January 2026.
- **Ms. Pratibha Dumane** delivered a talk on “Quality Assurance, Documentation and SDG Reporting Frameworks” on 20 June 2026 in the six day online Faculty Development Programme on “Transforming Institutions through Sustainability (SDGs), Quality Assurance and Impact - Driven Best Practices”, organized by the Nehru Institute of Engineering and Technology (Autonomous), Coimbatore, Tamil Nadu from 15 - 20 June 2026.



Faculty Development Programs (FDP)

- **Ms. Pratibha Dumane** attended a one-week ISTE-approved National Level Workshop on “Emerging significance of AI across Key Technological Domains such as Blockchain, IoT, Cybersecurity, Big Data Analytics, Cloud Computing and Robotics” organised from 1-7 July 2025 by VPM’s Maharshi Parshuram College of Engineering, Velneshwar.
- **Ms. Pratibha Dumane** completed one-day state-level online FDP on “Virtual Labs: Hands-on Outreach Workshop” organised by SNJB’s Shri Harilal Hastimal Polytechnic, Chandwad in association with Smt. Kashibai Navale College of Engg., Pune, the Institution of Engineers (India) Local Centre Nashik and IIT Bombay on 09 July 2025.
- **Dr. Aparna M. Telgote** attended a one-week AICTE Training and Learning (ATAL) Academy FDP on “Applications of Advanced Computing in Cyber Security and Forensic Science” organised by G. H. Raisoni College of Engineering and Management, Jalgaon from 18–23 August 2025.
- **Ms. Namita Agarwal and Ms. Pratibha Dumane** completed the one-week FDP on “Agentic AI (Building Autonomous Intelligent Systems)” organized by Innomatics Research Labs from 22-27 December 2025.
- **Ms. Namita Agarwal** attended a National Workshop on “IP Awareness and E-Patent Filing”, organized by VPM’s MPCOE, Velneshwar, in association with RGNIIPM, Nagpur & NIPAM, Intellectual Property India, Govt. of India on 16 January 2026.
- **Ms. Namita Agarwal** participated in IETE approved one-week FDP on “Communication Network Architecture and Security” organized by Lokmanya Tilak College of Engineering from 22-28 January 2026.
- **Ms. Pratibha Dumane** attended one-week online FDP on “AI for Sustainable Development” organized by the Dept. of Computer Engineering (Software Engineering), VIT Pune in association with IEEE Pune Section from 04-08 May 2026.
- **Ms. Namita Agarwal** attended one-week online FDP on “Recent Advances in AI: Machine Learning, Generative AI and Explainable Intelligence” organized by Rizvi College of Engineering, Mumbai from 29 June - 03 July 2026.

Certificate Courses by Faculty

1. **Ms. Poonam Chakraborty** completed the July - August 2025 NPTEL certificate course on “Innovation by Design”.
2. **Ms. Pratibha Dumane** completed the July-August 2025 NPTEL certificate course on “Technical Communication for Engineers” with an Elite + Silver award.
3. **Dr. Madhavi S. Pednekar** completed the August-October 2025 eight-week NPTEL online certificate course on “Yoga and Positive Psychology for Managing Career and Life”.
4. **Dr. Aparna M. Telgote** completed the following NPTEL courses:
    - “Data Science for Python” with an Elite award during July–August 2025.
    - “Enabling Technologies for 6G Communication Networks” with an “Elite + Silver” award during January–May 2026.
    - “Network Security” with an “Elite” award during January–May 2026.
5. **Ms. Namita Agarwal** completed the Jan-April 2026 NPTEL certificate course on “Computer Networks and Internet Protocol” with an “Elite + Silver” award. She is placed among the top 2% candidates certified for the course.
6. **Ms. Namita Agarwal** completed the SWAYAM certificate course on "Accreditation of Undergraduate Engineering Programme - As per SAR 2024-25” from January-June 2026 with a consolidated sco re of 93%.
7. **Mr. Ankur Ganorkar** completed the February - April 2026 NPTEL certificate course on “Fundamentals of Digital Marketing” with an “Elite + Silver” award.
8. **Ms. Freda Carvalho** completed the NPTEL certificate course on “Enabling Technologies for 6G Communication Networks” with a consolidated score of 94%. She is placed among the top 1% with Elite + Gold.
9. **Dr. Sudhakar Mande** attended the International VLSI Design & Embedded Systems Conference from 03-07 January 2026 at JW Marriott in Pune, Maharashtra.
10. **Dr. Sudhakar Mande, Dr. Madhavi Pednekar, Ms. Pratibha Dumane**, completed the 8 weeks, three credit SWAYAM online certificate course on "Accreditation of Undergraduate Engineering Programme - As per SAR 2024-25” from January-June 2026.
11. **Dr. Sudhakar Mande** completed the 4-week NPTEL online certificate course on “Product Design and Development” from July- August 2026.

Faculty Webinar Participation

- **Ms. Pratibha Dumane** attended the following webinars organised by Bharati Vidyapeeth's Institute of Computer Applications & Management (BVICAM), New Delhi:
  - Generative AI: Building End to End Agentic AI Applications on 06 December 2025.
  - Agentic AI: From LLM to Agents on 13 December 2025.
  - Beyond Spatial Audio: A Vision for the future of Human Connection through Immersive Calling on 17 January 2026.
  - Role of AI in 6G Networks on 14 February 2026.
  - Sustainable IT: Building Greener Future through Responsible Technologies on 21 February 2026.
  - Generative AI: Engineering, Deploying and Governing Responsible & Safe Agentic AI Systems on 07 March 2026.
  - Rethinking Document Processing in the Age of Agentic AI: From Documents to Decisions on 23 May 2026.
  - AI Under the Hood: Agents, Myths and the Art of Working with AI on 27 June 2026.
- **Ms. Namita Agarwal** attended a webinar on "Cyber Security – Tools to Monitor the Vulnerabilities" on 16 August 2025, organized by BVICAM, New Delhi.
- **Ms. Pratibha Dumane** attended "Let's Talk IP - IPR Awareness for Viksit Bharat" Program under the National Intellectual Property Awareness Mission (NIPAM) on 13 May 2026 held at RRU, Gandhinagar organized by Intellectual Property Office, India and Gujarat Council on Science and Technology (GUJCOST).

Faculty Engagements

1. Dr. Sudhakar Mande has been appointed as the Vice-Chancellor's Nominee for the Academic Council of the following colleges:

- Viva Institute of Technology, Vasai, Mumbai from AY 2026-27 to AY 2029-30.
- Vasantdada Patil Pratishthan's College of Engineering and Visual Arts, Sion, Mumbai from AY 2026-27 to AY 2029-30.
- Terna College of Engineering, Nerul, Navi Mumbai from AY 2026-27 to AY 2029-30.

2. Ms. Pratibha Dumane served as a Reviewer for the following international conferences:

- 5th International Conference on Advances and Applications of Artificial Intelligence and Machine Learning (ICAAAIML-2025) held on 18-19 July 2025, organised by Sharda School of Computing Science & Engineering, Sharda University, Greater Noida, Uttar Pradesh, India.
- International Conference on Advancing Technology, Engineering and Science (ICATES-2025) organised by Xavier Institute of Engineering, Mumbai in association with The Institution of Engineering & Technology, Mumbai Local Network, on 04-05 October 2025.
- 12th International Conference on Emerging Trends in Engineering & Technology Signal and Information Processing (ICETET SIP-25) on 01-02 August 2025 at G.H. Raisoni College of Engineering, Nagpur, India, technically sponsored by IEEE and co-sponsored by IEEE Bombay Section.
- AICTE-sponsored IEEE International Conference on Smart Technologies & Systems for Next Generation Computing (ICSTSN) 2025 - Fourth Edition held at IFET College of Engineering, Villupuram, India on 20-21 November 2025.
- 8th IEEE Pune Section International Conference IEEE PuneCon 2025 held from 12-15 December 2025, jointly organised by Savitribai Phule Pune University & IEEE Pune Section, Pune, India.

Faculty Engagements

3. Dr. Aparna Telgote was invited as a Session Chair for the International Conference on Smart Systems for Sustainable Development (SmartSSD-2026) on 15-16 April 2026 at Anjuman-I-Islam's M.H. Saboo Siddik College of Engineering, Mumbai.

4. Dr. Freda Carvalho served as a Reviewer for the journal "Progress in Electromagnetics Research" in March 2026.

5. Dr. Madhavi S. Pednekar was invited as a Reviewer for the following:

- 2nd IEEE International Conference on Information, Implementation, and Innovation in Technology 2026 (I2ITCON 2026), organized by Hope Foundation's International Institute of Information Technology, Pune on 30-31 July 2026 in Hybrid Mode.
- March-April Issue, International Journal of Innovative Research in Technology - IJIRT, ISSN: 2349-6002, Established: 2014, International Peer-Reviewed & Refereed Open Access Journal.

6. Dr. Aparna M. Telgote served as a Reviewer for the following:

- Journal e-Prime – Advances in Electrical Engineering, Electronics and Energy, published by Elsevier.
- Journal of Electrical Engineering and Technology, published by Springer Nature.
- "Smart Systems for Sustainable Development (SmartSSD-2026)", a conference organized by M. H. Saboo Siddik College of Engineering, held on 15–16 April 2026.
- 3rd International Conference on Technologies for Energy, Agriculture and Healthcare (ICTEAH 2026), held at K J Somaiya School of Engineering, Somaiya Vidyavihar University, Mumbai, on 16–17 April 2026.

7. Ms. Pratibha Dumane served as a "Core Member" in drafting the syllabus for the course "(PCC401) Neural Networks & Fuzzy Logic", Second Year, Electronics and Telecommunications Engineering, Semester IV (NEP 2020), University of Mumbai, and also attended the online faculty orientation programme for this course on 09 January 2026, organised by Thadomal Shahani Engineering College on behalf of University of Mumbai.

Faculty Engagements

8. Ms. Namita Agarwal served as the Chairperson for the Syllabus Drafting Committee for "Analog and Digital Communication" course under the University of Mumbai NEP 2020 scheme for Semester IV, Electronics and Telecommunication Engineering and conducted the Faculty Orientation Programme on 14 January 2026.

9. Ms. Poonam Chakraborty served as a Reviewer for the following conferences:

- 4th International Conference on Science, Technology, Engineering & Mathematics for Sustainable Development (ICSTEMSD-2026), organized by Atharva College of Engineering, Mumbai, on 24 March 2026.
- "Smart Systems for Sustainable Development (SmartSSD-2026)" organized by M. H. Saboo Siddik College of Engineering on 15–16 April 2026.

10. Dr. Sonal Patil served as a Reviewer for the Advanced Electromagnetics Journal.

11. Ms. Namita Agarwal served as a Judge for the Poster Presentation Competition organized by Fr. C. Rodrigues Institute of Technology, Vashi, Navi Mumbai on 02 May 2026.

12. Dr. Freda Carvalho was invited as the Chief Guest at Techfest at Smt. Jamnabai H. Wadhwa College of Technology on 13 February 2026.

IETE-DBIT-ISF

Core Team

Sr. No.	Office Bearers	Name
1	IETE-DBIT-ISF Faculty Coordinator	Mr. Ankur Ganorkar
2	Chairperson	Kartik Nyalapelli
3	Secretary	Daniel Sebastian
4	Treasurer	Sneha Pawar
5	Membership and Marketing Head	Blossom Mascarenhas
6	Technical Coordinators	Pratham Ghagre Vibhav Tripathi
7	Creative Design	Swanandi Chavan Tanvi Khamitkar
8	Membership & Marketing	Tanvi Shelar Abhinav Borse
9	Event & Logistics	Pushkar Mahale Shreyash Meher
10	Documentation	Nandana Anilkumar Sakshi Talkar

Activities by IETE-DBIT-ISF

A PCB Design Workshop using KiCad was organized on 01–02 April 2026 for First Year (Semester II) EXTC students to provide hands-on training in PCB design and the electronic circuit fabrication process. The sessions were conducted by EXTC-DBIT IETE ISF student members Mr. Vibhav Tripathi and Mr. Shreyash Meher.

The workshop was coordinated by Dr. Madhavi Pednekar and Mr. Ankur Ganorkar, while Mr. Deepak Pandav guided the participants through practical PCB design and fabrication concepts, helping students gain valuable industry-oriented skills.



Activities by IETE-DBIT-ISF

A visit to the Godrej K2 Plant, Khalapur, Navi Mumbai on 10 February 2026 was organised by Mr. Ankur Ganorkar for S.E. students.





IEEE-DBIT

Core Team



Sr. No.	Position	Name
1	Chairperson	Sanket Prajapati (TE MECH)
2	Vice-Chairperson	Shariya Ansari (TE IT)
3	Secretary	Moin Khan (TE EXTC)
4	Joint Secretary	Christoval D'souza (SE EXTC)
5	Treasurer	Aayush Jangam (TE EXTC)
6	Joint Treasurer	Amey Kanade (SE EXTC)
7	MTT-S Chair	Samiksha Gaonkar (TE EXTC)
8	MTT-S Vice Chair	Janhavi Shejwal (TE EXTC)
9	SIGHT Chair	Dhruv Tare (TE EXTC)
10	Joint SIGHT Chair	Vineet Suryavanshi (SE EXTC)

IEEE-DBIT Core Team

11	RAS Chair	Aryan Puranik (TE EXTC)
12	RAS Vice Chair	Neha Haldankar (SE EXTC)
13	Reporting	Aarya Jadhav (SE EXTC)
14	Joint Reporting	Shreya Manoj (FE EXTC)
15	Publicity	Tushita Patil (TE COMP)
16	Joint PR & Publicity	Deviprasad Shetty (SE IT)
17	WIE Chair	Shravani Pokale (TE COMP)
18	Joint WIE Chair	Ashlesha Tambat (SE EXTC)
19	Webmaster	Anay Deshpande (TE COMP)
20	Joint Webmaster	Sanket Jadhav (SE COMP)
21	CS Chair	Smruti Alegaonkar (TE IT)
22	CS Vice Chair	Jaeden Pereira (TE COMP)



Activities by IEEE-DBIT



KiCAD Workshop



Django Workshop



Rocket League

Activities by IEEE-DBIT

- IEEE DBIT CS Student Chapter organized “Open-Source Contribution Workshop” on 23 July 2025 for students of all departments.

- IEEE DBIT CS Student Chapter organized “Hugo Static Website Generator Workshop” on 30 July 2025 for students of all departments.

- The KiCAD Workshop organised by IEEE DBIT MTT-S Chapter, held on 13 August 2025, by Mr. Premkumar Singh (Alumnus DBIT), Trainee Engineer, Selec Controls was conducted for SE EXTC students.

- IEEE DBIT RAS conducted an Arduino Hands-on Workshop on 09 December 2025 to introduce participants to the fundamentals of Arduino hardware and embedded programming.

- The IEEE DBIT CS hosted a workshop on Small Language Models (SLMs) on 24 February 2026. The session was led by the resource person - Mr. Lucky Rathore, Founder, Sci-fi.

- The Full-Stack Django Web Development Workshop, held on 05-06 March 2026, was organized by the IEEE DBIT Computer Society Chapter in collaboration with Colosseum 2026.

- The Rocket League competition, organised by the IEEE DBIT Core Team under Colosseum 2026, was an innovative and engaging event that combined gaming concepts with real-world robotics.

- IEEE MTT-S Student Chapter in association with IEEE-DBIT organized a session on “Radar and Automotive Radar” on 11 April 2026 by Mr. Jason Fernandes, Radar Expert at BMW AG.

ColoHacks : Hardware Hackathon

The 24-hour Hardware Hackathon organised by IEEE DBIT under Colosseum 2026 was a highly intensive and innovation-driven event conducted on 24-25 March 2026, bringing together participants from multiple colleges to design and develop real-time hardware solutions addressing practical, real-world problems. A total of 23 teams were shortlisted for the final offline hackathon.





Class Photographs

Final Year



Third Year



EXTC Department Educators



The Network Behind the Tech

Student Editorial Team



Pushkar Mahale



Swanandi Chavan



Saksham More



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